# 1 Hypothesis

# 2 Goal

- Make a story.
- Connect all the prototypes.
- Improve the prototype algorithm.

# 3 Story

The player will find themselves in a sort of laboratory where a disembodied voice will try to guide the player out of the laboratory. If the player follows the voice they will successfully escape the laboratory. If the player disobeys the voice they will be taken deeper into the laboratory where they will then have to escape from the laboratory without the help of the narrator.

# 4 Game Progression

The general game progression is that the player will start in the tutorial and then make their ways from the first level to the third level. The tutorial and first level will be handmade and will have a narrator on both levels. The second and third level will both be procedurally generated with different values in their generation to indicate a decay of the laboratory.

## 4.1 Tutorial

The player will start in the tutorial where they will learn the basics of the game. The player will first learn the basic controls which are: Moving, sprinting, toggling the player light and general interactions the player can perform. The player will be introduced to the enemy in the tutorial and different ways to avoid them.

## 4.2 First Level

The player will be put in a laboratory map where a disembodied voice called the narrator will then try to help the player escape from the laboratory. If the player follows the narrator then they will escape from the laboratory, if the player disobeys the narrator they will go deeper into the laboratory. The enemy will not appear in this level. This level will not be procedurally generated.

### 4.3 Second Level

The player will be in a laboratory that is starting to decay. The player will then need to complete a puzzle while hiding and avoiding the enemy. After the player completes the puzzle they will be taken to the next level through either a door opening on the map or being teleported there. The decay will be shown by giving some random walk to the procedural generation.

### 4.4 Third Level

The player will be in a laboratory that is decayed. The player will then face the same challenges that they faced in the second level. The decay is shown by giving the procedural generation a high level of random walk.

## 5 Genre

The genre of the game will be a stealth survival horror and puzzle genre. The player will need to hide and evade an unstoppable enemy while trying to complete some sort of goal. This will be similar to the Alien from Alien Isolation, because the AI enemy was inspired from Alien Isolation. The puzzle genre appears when the player will need to find solutions to the puzzles presented to them.

## 6 Possible Puzzles

These are the puzzles that can appear during the procedural generation of the level. Each puzzle will have success criteria which is determined by the puzzle.

### 6.1 Fetch quest

The player will need to find an item and bring it to the destination.
The generation will be a success if:

- The item can be accessed.
- The item can be picked up and put down.
- The destination can be accessed.
- The destination recognizes the item.
- The player completes the goal after brining the item to the destination.

### 6.2 Password

The player will need to find a password that they will then use to open a lock. The generation will be a success if:

- All parts of the password can be accessed
- All parts of the password can be read.
- The lock can be accessed.
- The lock successfully recognizes the entered password.
- The lock completes the goal after the password has been entered.

### 6.3 Pattern

The player will be presented with a special room in which the player will need to move objects in a specific order to complete the puzzle. The generation will be a success if:

- All parts of the puzzle can be accessed
- All parts of the puzzle can be picked up and put down.
- The puzzle can be accessed.
- The puzzle successfully recognizes the entered pattern.
- The puzzle completes the goal after the pattern has been entered.

## 7 Prototype Improvements

This prototype will improve upon all previous prototypes, while connecting all the prototypes together.

### 7.1 Prototype 1 Improvements

The map of prototype 1 will be improved while also allowing the player to use more of the mechanics to explore the map. This will include areas that can only be accessed when the player's light is turned off and areas that have a timer that the player needs to sprint towards to access. This will give the player more interaction in exploring the map.

### 7.2 Prototype 2 Improvements

The tutorial will include the narrator to help the player understand the story of the game. This will also include improving the map to allow the player to improve on the mechanics while learning them. This level will also require an improved enemy to show the player the actions and behavior of the enemy.

### 7.3 Prototype 3 Improvements

This prototype focused on the enemy thus the enemy will be improved. The enemy will be given a static state in which they will not move. In this static state the enemy will only listen for player footsteps and move towards them. The enemy sight will also be shown in the tutorial to the player in this state to help the prototype 2 improvements. This new state can get the player to possible stun the enemy allowing for ways to fight against but not remove the enemy.

### 7.4 Prototype 4 Improvements

The procedural generation algorithm will be improved by keeping track of all the rooms in the generation to allow adding of special rooms which will either be a room in which a puzzle needs to be solved or will have an interaction for the player. The algorithm will also allow for secret passages between rooms.

## 8 Communication Design

## 9 Process